

Nifty 100 Financial Analytics Platform — Analyst Guide

Comprehensive User Manual & Technical Documentation

1. Introduction & Platform Architecture

The Nifty 100 Financial Platform is designed for institutional equity research, offering automated financial statement parsing, ratio computation, NLP sentiment analysis, KMeans clustering, and REST API distribution.

Detailed technical steps, parameter definitions, and usage guidelines are provided for analysts.

2. Navigating the Streamlit Dashboard

The dashboard features 5 core pages: Executive Summary, Company Profile, Screener & Filter, Peer Comparison, and Data Quality Monitor.

Detailed technical steps, parameter definitions, and usage guidelines are provided for analysts.

3. Using the Equity Screener

Filter companies across 10 core financial ratios including ROE, D/E, FCF, and 5-Year Revenue CAGR. Presets include Quality Compounders and Value Opportunities.

Detailed technical steps, parameter definitions, and usage guidelines are provided for analysts.

4. Company Profile & Tearsheet Generation

Each profile provides 10-year P&L, Balance Sheet, and Cash Flow trajectories alongside NLP Pros & Cons and automated PDF tearsheets.

Detailed technical steps, parameter definitions, and usage guidelines are provided for analysts.

5. Peer Comparison & Radar Analytics

Benchmark target companies against sector median and peer group percentiles using radar metric charts.

Detailed technical steps, parameter definitions, and usage guidelines are provided for analysts.

6. Machine Learning Clustering (KMeans)

Companies are segmented into 5 distinct archetypes: High-Quality Compounders, Defensive Dividend Payers, Value Cyclicals, Distressed Turnarounds, and Emerging Growth.

Detailed technical steps, parameter definitions, and usage guidelines are provided for analysts.

7. Cash Flow Intelligence & Capital Allocation

Evaluates CFO/EBITDA quality scores, CapEx intensity, and flags capital allocation pattern changes.

Detailed technical steps, parameter definitions, and usage guidelines are provided for analysts.

8. Data Quality & Exception Management

Monitors 14 automated data quality rules, outputting validation failures and severity tags.

Detailed technical steps, parameter definitions, and usage guidelines are provided for analysts.

9. REST API Integration & Endpoints

FastAPI server exposes 16 endpoints for integration with external platforms under `/api/v1` prefix.

Detailed technical steps, parameter definitions, and usage guidelines are provided for analysts.

10. Troubleshooting & System Maintenance

Instructions for database re-indexing, cache clearing, and running the automated pytest validation suite.

Detailed technical steps, parameter definitions, and usage guidelines are provided for analysts.